

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Frequency domain sampling theorem

The spectral $X(e^{j\Omega})$ characteristics of the discrete non periodic sequence $x[k]$:

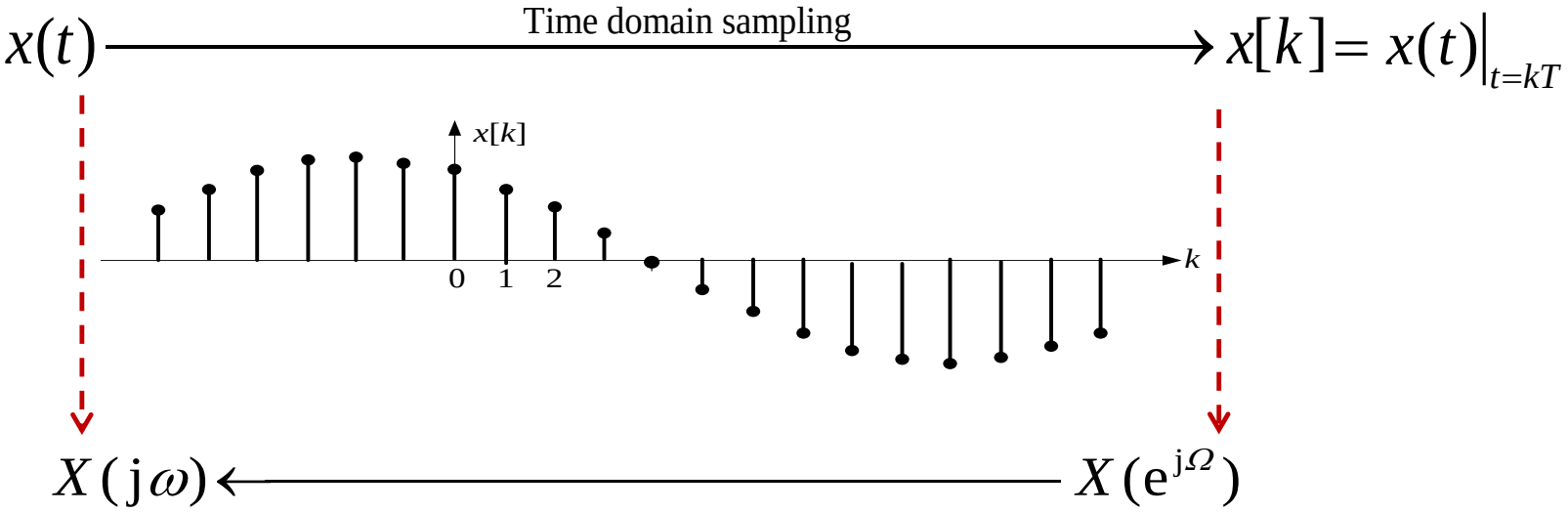
(1) $X(e^{j\Omega})$ is a continuous function of Ω

(2) $X(e^{j\Omega})$ is a periodic function with a period of 2π

In order to use numerical methods to calculate the spectrum $X(e^{j\Omega})$, it is necessary to analyze $X(e^{j\Omega})$

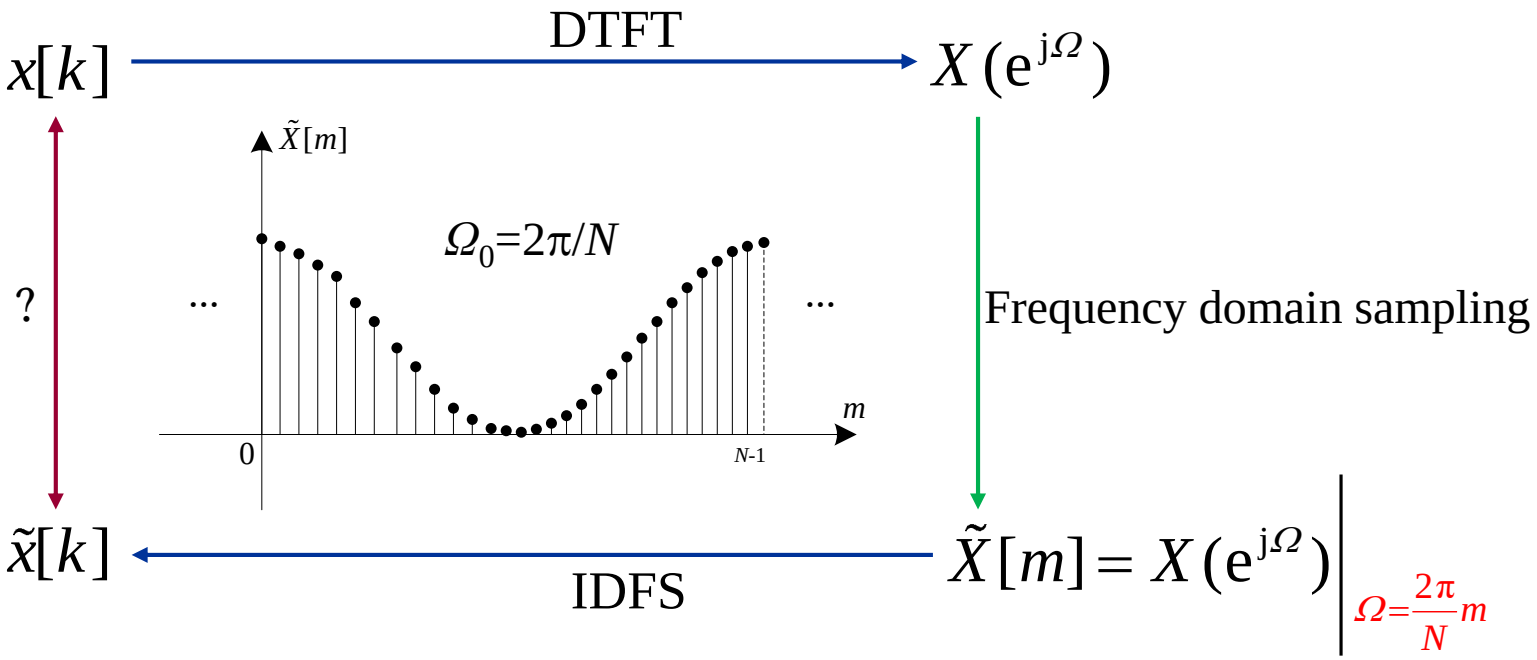
Perform frequency domain sampling to obtain its corresponding discrete spectrum.

Frequency domain sampling theorem



$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[j(\omega - n\omega_{sam})\right] \qquad (\Omega = \omega T)$$

Frequency domain sampling theorem



Frequency domain sampling theorem

Frequency domain sampling theorem:

If $\tilde{X}[m]$ is Periodic sequence of $X(e^{j\Omega})$ at equidistant ($\Omega_0=2\pi/N$) sampling points

$$\tilde{X}[m]=X(e^{j\Omega})\Big|_{\Omega=\frac{2\pi}{N}m} \qquad m=0,1,\cdots,N-1$$

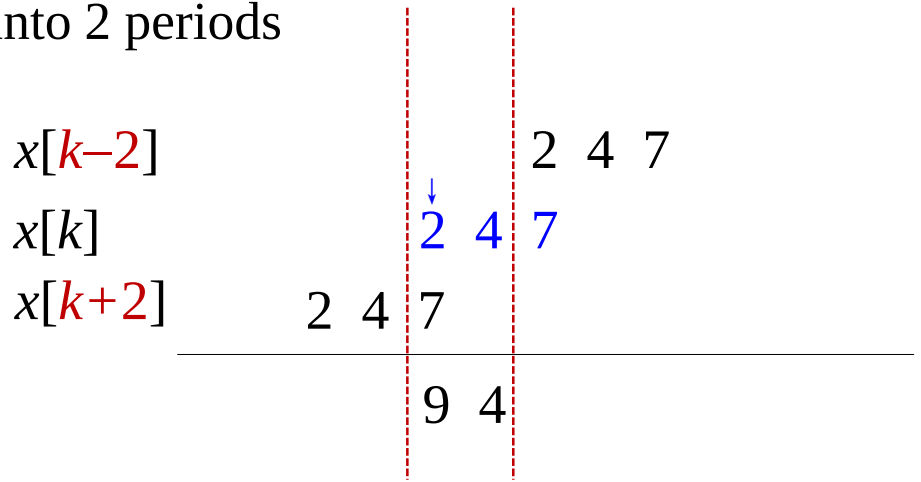
$$\tilde{x}[k]=\mathrm{IDFS}\{\tilde{X}[m]\}=\sum_{l=-\infty}^{\infty}x[k+lN]$$

The discretization of the frequency spectrum $X(e^{j\Omega})$ of sequence $x[k]$ corresponds to the periodicity of its time-domain sequence $x[k]$ with a period of N.

[E.g] Given that the frequency spectrum of a finite sequence $x[k]=\{2, 4, 7;\ k=0,1,2\}$ is $X(e^{j\Omega})$, Sampling X $(e^{j\Omega})$ at N=2, 3, and 4 points within one cycle, IDFS$\{\tilde{X}[m]\}$?

Solution: According to the frequency domain sampling theorem,

The time-domain sequence $x[k]$ is divided into 2 periods

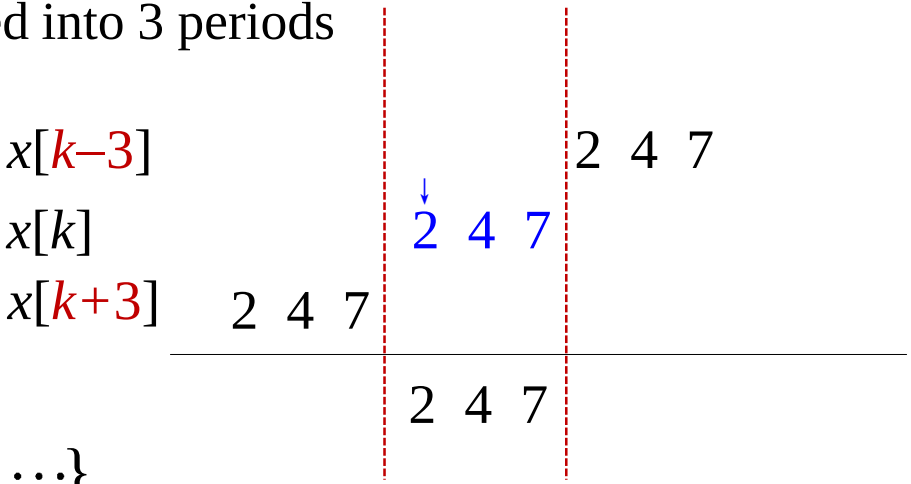


$$\text{IDFS}\{\tilde{X}[m]\} = \tilde{x}_2[k] = \{\cdots, \underset{\downarrow}{9}, 4, \cdots\}$$

[E.g] Given that the frequency spectrum of a finite sequence $x[k]=\{2, 4, 7;\ k=0,1,2\}$ is $X(e^{j\Omega})$, Sampling X $(e^{j\Omega})$ at N=2, 3, and 4 points within one cycle, IDFS$\{\tilde{X}[m]\}$?

Solution: According to the frequency domain sampling theorem,

The time-domain sequence $x[k]$ is divided into 3 periods



$$\text{IDFS}\{\tilde{X}[m]\} = \tilde{x}_2[k] = \{\cdots, \underset{\downarrow}{2}, 4, 7, \cdots\}$$

[E.g] Given that the frequency spectrum of a finite sequence $x[k]=\{2, 4, 7;\ k=0,1,2\}$ is $X(e^{j\Omega})$, Sampling X $(e^{j\Omega})$ at N=2, 3, and 4 points within one cycle, IDFS$\{\tilde{X}[m]\}$?

Solution: According to the frequency domain sampling theorem,

The time-domain sequence $x[k]$ is divided into 2 periods

$x[k-4]$ 2 4 7 0

$x[k]$ ↓ 2 4 7 0

$x[k+4]$ 2 4 7 0

2 4 7 0

$$\text{IDFS}\{\tilde{X}[m]\}=\tilde{x}_2[k]=\{\cdots,\underset{\downarrow}{2},4,7,0,\cdots\}$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.